

M109

تدريبات العملي



إن الله لا يخيب عبداً سعى

Keep going sometimes you win sometimes you learn

"Prepared by Aseela"

TMA PLT revision

Q1: Develop a C# code that will ask the user to input the number of people that want to buy tickets of the coming football match, and input the category of the ticket based on the following:

1)VIP

2)IP

3)P

Finally, the code will calculate and display the total of the tickets keeping in mind that the price of the tickets is as follows:

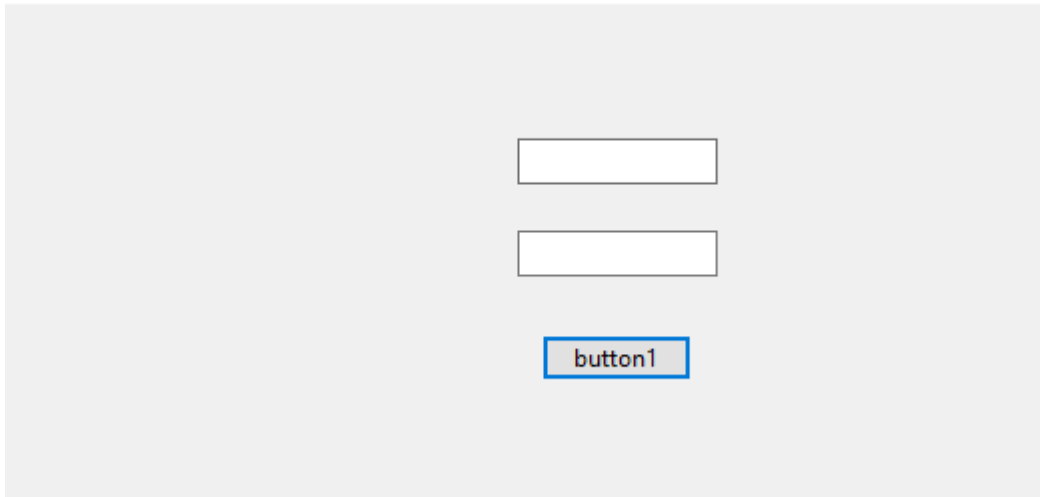
1)VIP = 500 Riyals

2)Ip = 300 Riyals

3)P = 100 Riyal.

واجهة الForm :

Form1



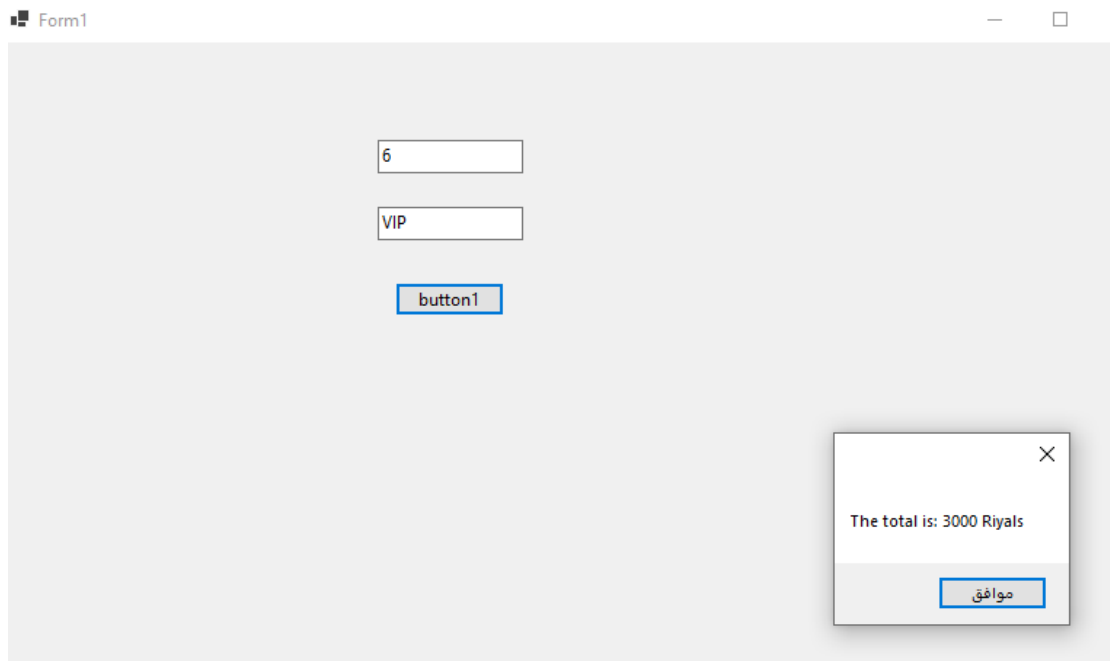
الكود:

```
private void button1_Click(object sender, EventArgs e)
{
    int n = int.Parse(textBox1.Text);
    string m = textBox2.Text;
    int total = 0;

    if (m == "VIP")
    {
        total = 500 * n;
    }
    else if (m == "IP")
    {
        total = 300 * n;
    }
    else if (m == "P")
    {
        total = 100 * n;
    }
    else
    {
        MessageBox.Show("Invalid input, enter only VIP, IP or P");
    }

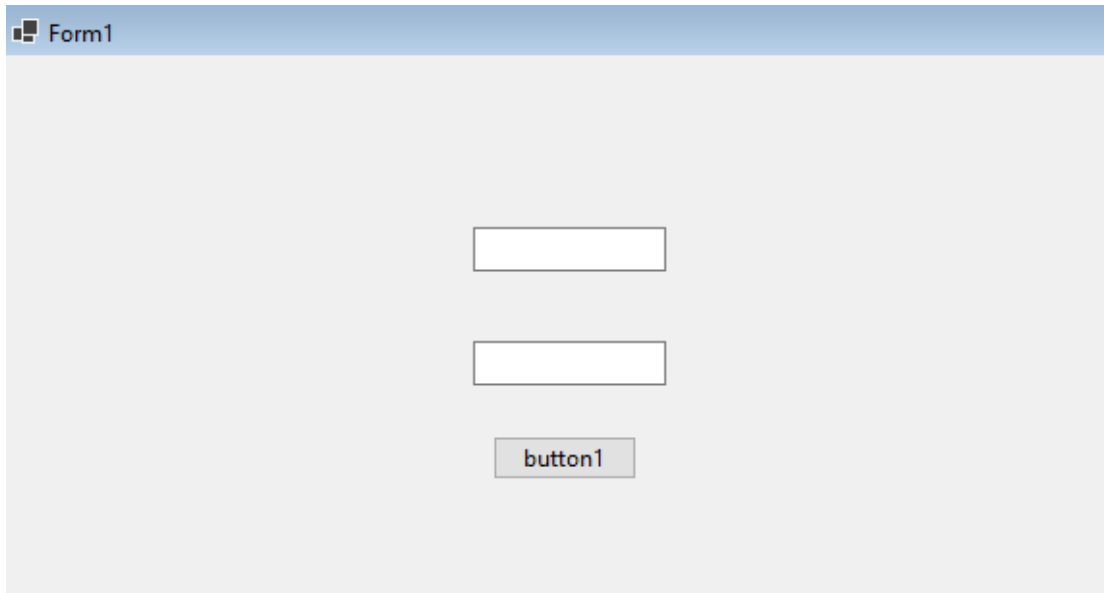
    MessageBox.Show("The total is: " + total + " Riyals");
}
```

The output:



Q2: Develop C# code that will ask the user to input a starting and ending value, then these two numbers will be passed to lamsumofeven method, the method will calculate the sum of all even numbers from the starting to the end values, finally the sum will be displayed

واجهة ال Form:



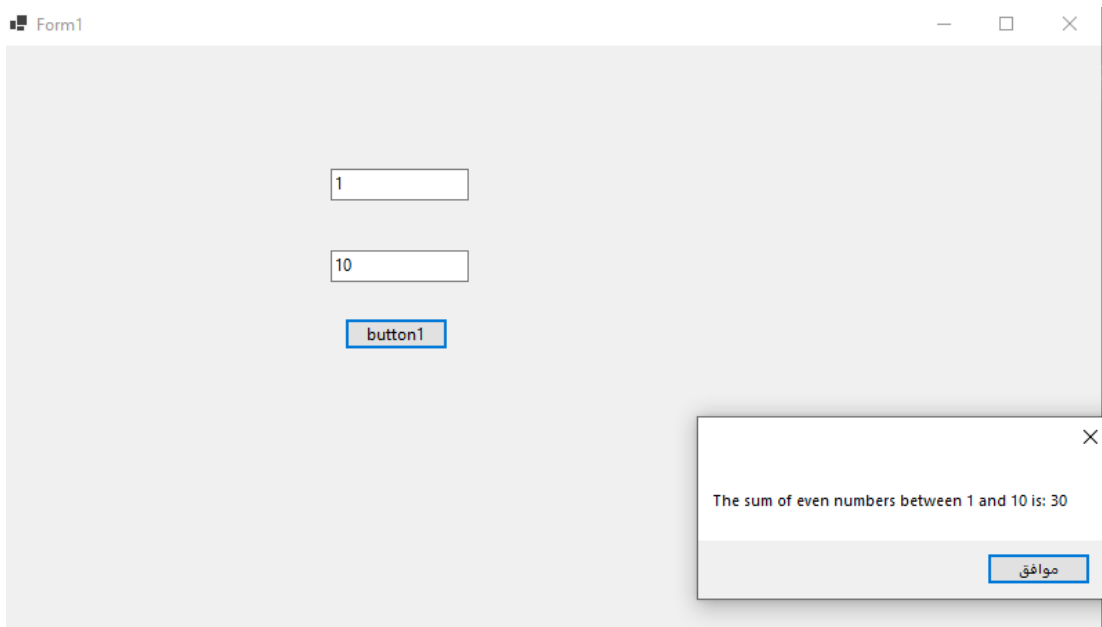
The image shows a Windows Form titled "Form1". It has a light gray background. In the center, there are two white text boxes with black borders, one above the other. Below the text boxes is a button with a gray border and the text "button1" in the center.

الكود:

```
private void button1_Click(object sender, EventArgs e)
{
    int c = int.Parse(textBox1.Text);
    int d = int.Parse(textBox2.Text);
    lamsumofeven(c, d);
}

1 reference
private void lamsumofeven(int x, int y)
{
    int sum = 0;
    for (int i = x; i <= y; i++)
    {
        if (i % 2 == 0)
        {
            sum += i;
        }
    }
    MessageBox.Show("The sum of even numbers between " + x + " and " + y + " is: " + sum);
}
```

The output:



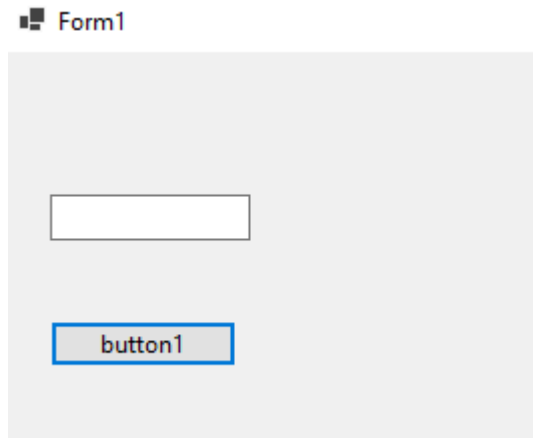
**Q3: Develop a C# program that will have the following array:
{Ahmed, Dalal, Saleh, Tala}**

Then the code will ask the user to input the name we are looking for, and the code will search for that name in the array, if the name is found then the following will be displayed:

- 1) The name we are searching and find in the array.**
- 2) The number of the position of the name in the array.**
- 3)The number of the index of the name in the array.**

Otherwise, the code will display a sorry message for not finding the name.

واجهة ال Form:



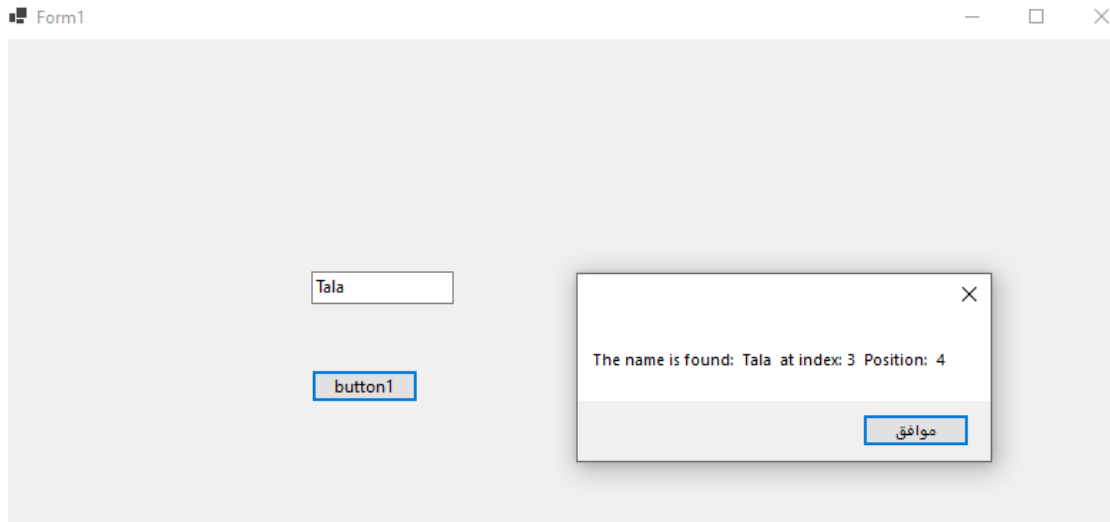
الكود:

```
private void button1_Click(object sender, EventArgs e)
{
    string[] names = { "Ahmed", "Dala", "Salah", "Tala"};
    string s=textBox1.Text;

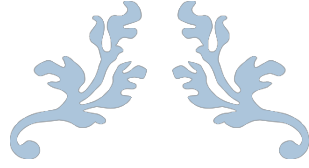
    if (names.Contains(s))
    {
        int i = Array.IndexOf(names,s);
        int p = i + 1;
        MessageBox.Show("The name is found" + s + " at index: " + i + "Position: " + p);
    }
    else
    {
        MessageBox.Show("The name is not found in the array.");
    }
}

}
```

The output:



سبحان الله وبحمده.....سبحان الله العظيم



M109

تدريبات للعملي بارت 2



Q1: Develop a C# application to determine the cost of museum ticket.
The application should prompt the user for the kind of museum:

1. Children Museum
2. Art Museum
3. History Museum
4. Science Museum

Then the application should ask whether the user is in the reward program or not. If the user is in the reward program, a 5 S.R deduction is applied. Finally, the application should calculate the cost keeping in mind that the price of each kind of museum is as follows:

1. Children's Museum = 10 S.R
2. Art Museum = 15 S.R
3. History Museum = 30 S.R
4. Science Museum = 45 S.R

شكل الForm:



The image shows a simple Windows Forms application window with a light gray background. It contains three main UI elements arranged vertically: a text box at the top, a checkbox with the Arabic label 'عندي مكافئة' (I have a reward) in the middle, and a button labeled 'button1' at the bottom.

الكود:

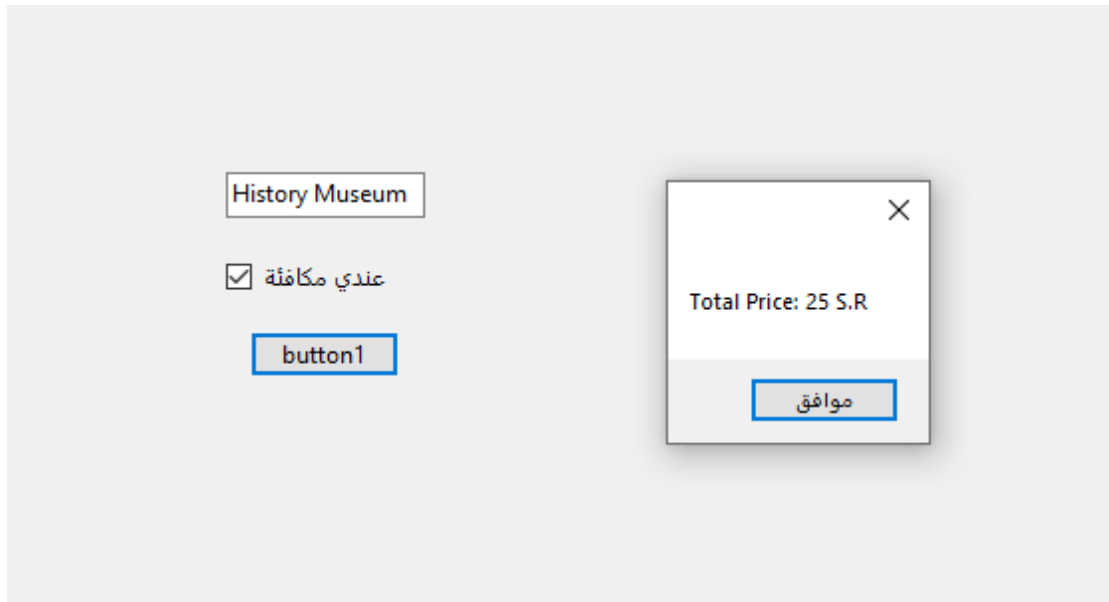
```
private void button1_Click(object sender, EventArgs e)
{
    int p = 0;
    string m=textBox1.Text;

    if (m == "Children Museum")
        p = 10;
    else if (m == "Art Museum")
        p = 15;
    else if (m == "History Museum")
        p = 30;
    else if (m == "Science Museum")
        p = 45;

    if (checkBox1.Checked)
    {
        p=p-5;
        MessageBox.Show("Total Price: " + p+ " S.R");
    }

    else
    {
        MessageBox.Show("Total Price: " + p + " S.R (No Discount)");
    }
}
```

The output:

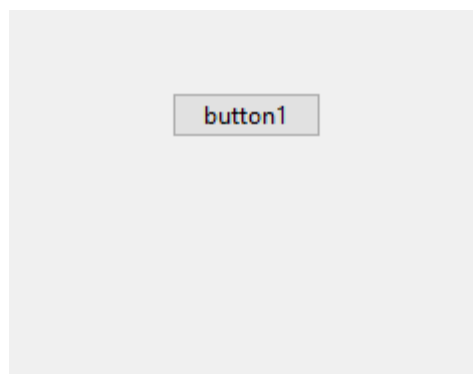


Q2: Develop a C# program that will have the following array:

$\{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$

Then the code should iterate through the arrays and multiply all numbers and display the product.

شكل ال Form:



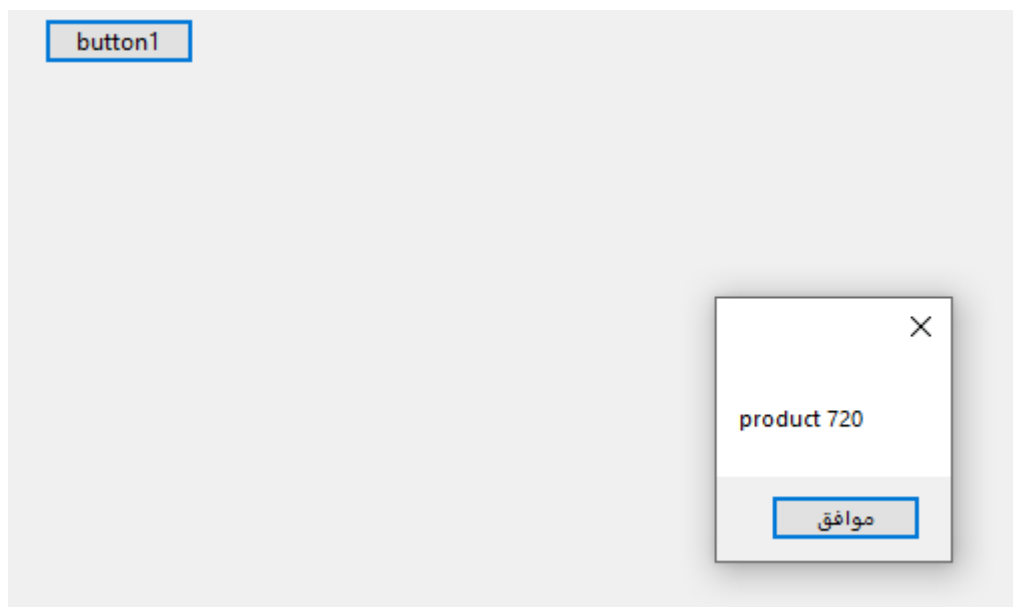
الكود:

```
private void button1_Click(object sender, EventArgs e)
{
    int[,] s = { { 1, 2 },
                 { 3, 4 },
                 { 5, 6 } };

    int product = 1;
    for (int i = 0; i < s.GetLength(0); i++)
        for (int j = 0; j < s.GetLength(1); j++)
            product = product * s[i, j];

    MessageBox.Show("product "+product);
}
```

The output:



Q3: Develop a C# program that accepts a value between 0 – 100 from the user and then displays the product quality based on the following rating system:

- 0 – 20 : Poor
- 21 – 40 : Fair
- 41 – 60: Average
- 61 – 80: Very Good
- 81 – 100 : Excellent

An appropriate message should be displayed if an invalid value is entered.

شكل الForm:

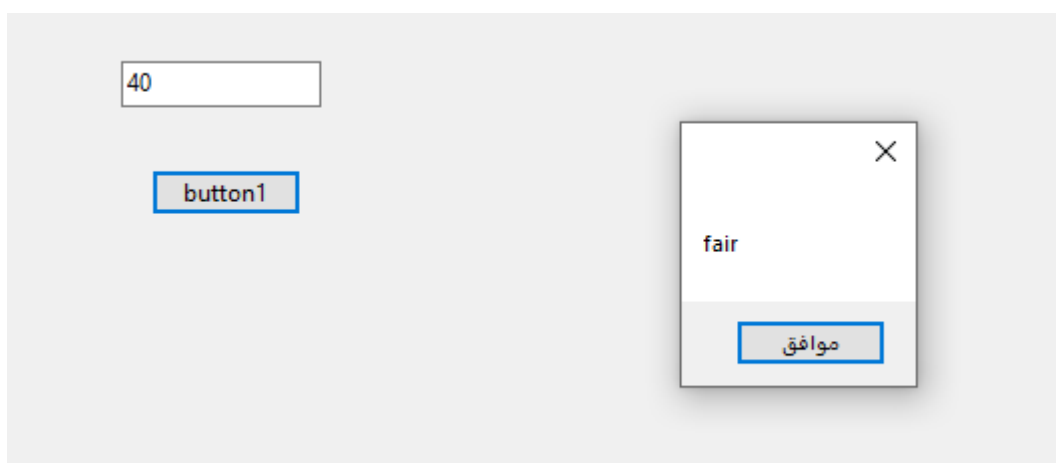


The image shows a simple Windows Form. It has a light gray background. In the center, there is a white text input box. Below the input box, there is a button with a gray border and the text 'button1' inside it.

الكود:

```
private void button1_Click(object sender, EventArgs e)
{
    if (int.TryParse(textBox1.Text, out int v))
    {
        if (v >= 0 && v <= 100)
        {
            string m = "";
            if (v <= 20)
                m = "poor";
            else if (v <= 40)
                m = "fair";
            else if (v <= 60)
                m = "good";
            else if (v <= 80)
                m = "very good";
            else
                m = "excellent";
            MessageBox.Show(m.ToString());
        }
        else
        {
            MessageBox.Show("Please enter a valid number between 0 and 100.");
        }
    }
}
```

The output:



لا حول ولا قوة إلا بالله

M109

TMA PLT revision 3

Q1: Develop a C# program that prompts the user to enter the number of hours. The program should convert the number of hours into its equivalent minutes and seconds. An appropriate message should be displayed if the entered number is not positive.

One hour = 60 minutes

One hour = 3600 seconds

شكل الفورم:




```

1 reference
private void button1_Click(object sender, EventArgs e)
{
    if (int.TryParse(textBox1.Text, out int h))
    {
        if (h > 0 && h <= 12)
        {
            int m = h * 60;
            int s = h * 60 * 60;

            label1.Text = ("One hour = " + m + " minutes ");
            label2.Text = ("One hour = " + s + " seconds");
        }
        else
        {
            label1.Text = "Please enter a valid hour (1-12)";
            label2.Text = " ";
        }
    }
    else
    {
        label1.Text = "Please enter a valid number";
        label2.Text = " ";
    }
}

```

الكود:

1

button1

One hour = 60 minutes

One hour = 3600 seconds

The output:

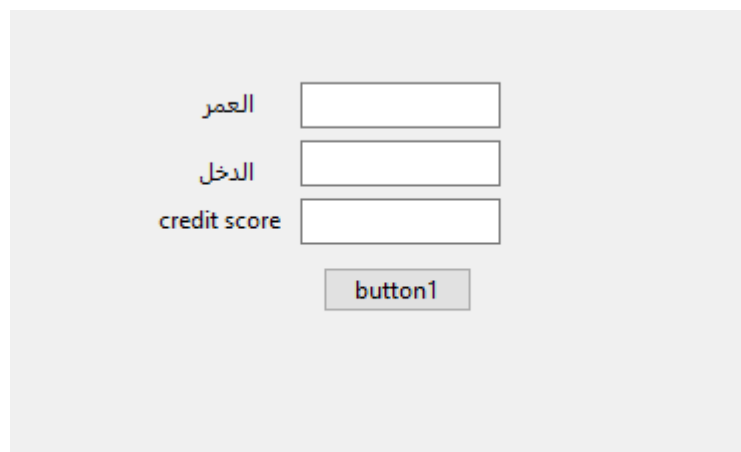
Q2: Develop a C# program that checks if the user is applicable, applicable with conditions or not applicable to obtain a loan. The program should prompt the user's age. If the age is greater than or equal 25 years old, then the program must prompt the user's income and credit score. A message must be displayed based on the following:

If income is higher than 50,000 S.R and Credit Score is higher than 700 S.R, then the user is applicable.

If income is higher than 30,000 S.R and Credit Score is higher than 600 S.R, then the user is applicable with conditions.

Otherwise, the user is not applicable to obtain a loan.

شكل الفورم:



الكود:

```

private void button1_Click(object sender, EventArgs e)
{
    if (int.TryParse(textBox1.Text, out int a))
    {
        if (a >= 25)
        {
            if (int.TryParse(textBox2.Text, out int c))
            {
                if (int.TryParse(textBox3.Text, out int s))
                {
                    if( c > 50000 && s > 700 )
                        MessageBox.Show("Applicable for loan");
                    else if( c > 30000 && s > 600 )
                        MessageBox.Show("Applicable for comditions");
                    else
                        MessageBox.Show("Not applicable for loan");
                }
            }
            else
            {
                MessageBox.Show("Please enter a valid credit card score.");
            }
        }
        else
        {
            MessageBox.Show("Please enter a valid income.");
        }
    }
    else
    {
        MessageBox.Show("age less than 25");
    }
}
}
else
{
    MessageBox.Show("Please enter a valid age.");
}
}
}

```

The output:

The screenshot displays a Windows application interface. On the left, there is a form with three text input fields. The first field, labeled 'العمر' (Age), contains the value '25'. The second field, labeled 'الدخل' (Income), contains '51000'. The third field, labeled 'credit score', contains '701'. Below these fields is a button labeled 'button1'. To the right of the form, a modal dialog box is open. The dialog box has a title bar with a close button (X) and the text 'Applicable for loan'. Inside the dialog, there is a single button labeled 'موافق' (Agree).

Q3: Develop a C# program that accepts an array from the user and then displays the elements of the array in a reversed order

الفورم:

الكود:

```

// Reference
private void button1_Click(object sender, EventArgs e)
{
    // نقرأ الأرقام من التيكست بوكس
    string x = textBox1.Text;
    // ننشئ مصفوفة ونقسمها لأجزاء بالمسافة
    string[] l = x.Split(' ');
    // نجهز مصفوفة أرقام بنفس الطول
    int[] y = new int[l.Length];

    // نحول النصوص لأرقام
    for (int i = 0; i < l.Length; i++)
    {
        y[i] = int.Parse(l[i]);
    }

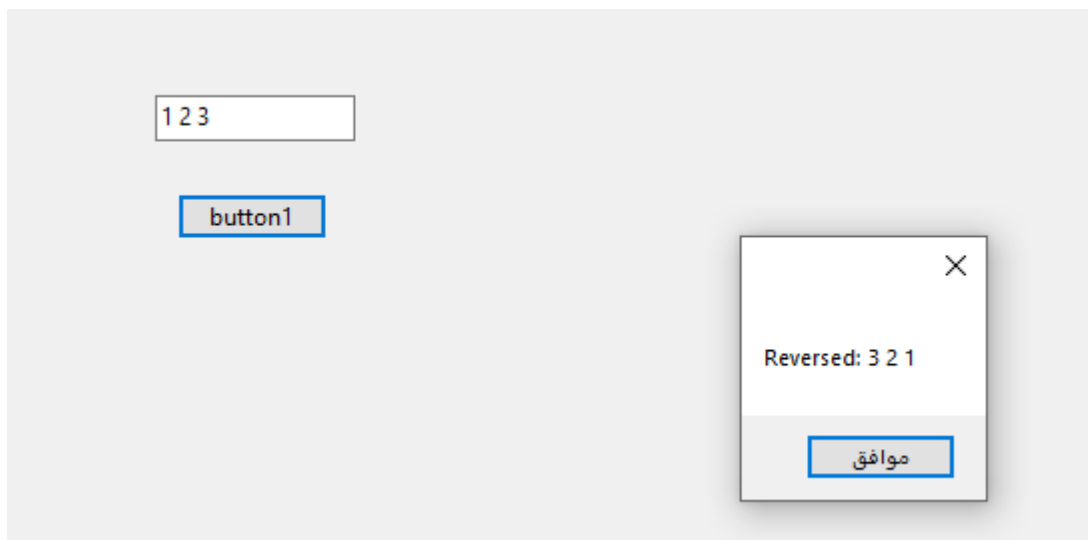
    // نعكس المصفوفة
    Array.Reverse(y);

    // نحول الأرقام لنص لعرضها
    string r = "";
    for (int i = 0; i < y.Length; i++)
    {
        r += y[i] + " ";
    }

    // نعرض النتيجة
    MessageBox.Show("Reversed: " + r);
}

```

The output:



أستغفر الله

C# Array Practice Examples with Solutions

Sum of Array Elements

Description: Ask the user to enter numbers separated by spaces, store them in an array, then calculate and display the sum of the elements.

Example Input: 5 10 15

Expected Output: Sum: 30

Code:

```
string input = textBox1.Text;
string[] parts = input.Split(' ');
int[] numbers = new int[parts.Length];
int sum = 0;

for (int i = 0; i < parts.Length; i++)
{
    numbers[i] = int.Parse(parts[i]);
    sum += numbers[i];
}

MessageBox.Show("Sum: " + sum);
```

sum

Find Maximum Value

Description: Ask the user to enter numbers separated by spaces, store them in an array, and then display the maximum value.

Example Input: 8 3 15 22 7

Expected Output: Max: 22

Code:

```
string input = textBox1.Text;
string[] parts = input.Split(' ');
int[] numbers = new int[parts.Length];

for (int i = 0; i < parts.Length; i++)
{
    numbers[i] = int.Parse(parts[i]);
}
```

بجای یه حلقه الی

لو طلب ال Min
نقسمها بس خط
Min();

```
int max = numbers.Max();  
MessageBox.Show("Max: " + max);
```

Count Even Numbers

Description: Ask the user to enter numbers separated by spaces, store them in an array, and count how many of them are even.

Example Input: 2 5 8 11 14

Expected Output: Even Count: 3

Code:

```
string input = textBox1.Text;  
string[] parts = input.Split(' ');  
int[] numbers = new int[parts.Length];  
int evenCount = 0;  
  
for (int i = 0; i < parts.Length; i++)  
{  
    numbers[i] = int.Parse(parts[i]);  
    if (numbers[i] % 2 == 0)  
        evenCount++;  
}  
  
MessageBox.Show("Even Count: " + evenCount);
```

لو جلب even و odd

```
private void button1_Click(object sender, EventArgs e)  
{  
    // نقرأ الأرقام من التيكست بوكس  
    string x = textBox1.Text;  
    // نلحق مصفوفة ونقسمها لأجزاء بالمعقافه  
    string[] l = x.Split(' ');  
    // نجهز مصفوفة أرقام بنفس الطول  
    int[] y = new int[l.Length];  
    int ce = 0;  
    int co = 0;  
  
    // نحول النصوص لأرقام  
    for (int i = 0; i < l.Length; i++)  
    {  
        y[i] = int.Parse(l[i]);  
        if (y[i] % 2 == 0)  
            ce++;  
        else  
            co++;  
    }  
  
    // نعرض النتيجة  
    MessageBox.Show("count of even is: " + ce + " odd is: " + co);  
}
```

Check for Specific Number

Description: Ask the user to enter numbers separated by spaces, then check if the number 10 is in the array and display a message.

Example Input: 1 3 10 20

Expected Output: 10 is in the array

Code:

```
string input = textBox1.Text;  
string[] parts = input.Split(' ');  
int[] numbers = new int[parts.Length];  
bool found = false;  
  
for (int i = 0; i < parts.Length; i++)  
{  
    numbers[i] = int.Parse(parts[i]);  
    if (numbers[i] == 10)  
        found = true;  
}
```

```

        found = true;
    }

    if (found)
        MessageBox.Show("10 is in the array");
    else
        MessageBox.Show("10 is not in the array");
}

```

Reverse Array

Description: Ask the user to enter numbers separated by spaces, store them in an array, then reverse and display them.

Example Input: 5 10 15

Expected Output: Reversed: 15 10 5

Code:

```

string input = textBox1.Text;
string[] parts = input.Split(' ');
int[] numbers = new int[parts.Length];

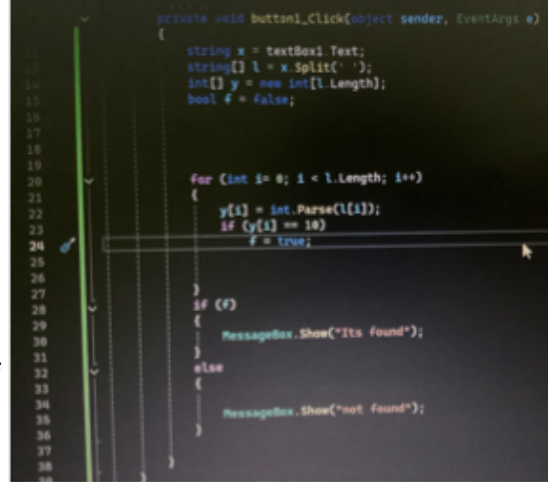
for (int i = 0; i < parts.Length; i++)
{
    numbers[i] = int.Parse(parts[i]);
}

Array.Reverse(numbers);

string result = "";
for (int i = 0; i < numbers.Length; i++)
{
    result += numbers[i] + " ";
}

MessageBox.Show("Reversed: " + result);

```



```

private void Button1_Click(object sender, EventArgs e)
{
    string x = textBox1.Text;
    string[] l = x.Split(' ');
    int[] y = new int[l.Length];
    bool f = false;

    for (int i = 0; i < l.Length; i++)
    {
        y[i] = int.Parse(l[i]);
        if (y[i] == 10)
            f = true;
    }
    if (!f)
    {
        MessageBox.Show("Its found");
    }
    else
    {
        MessageBox.Show("not found");
    }
}

```